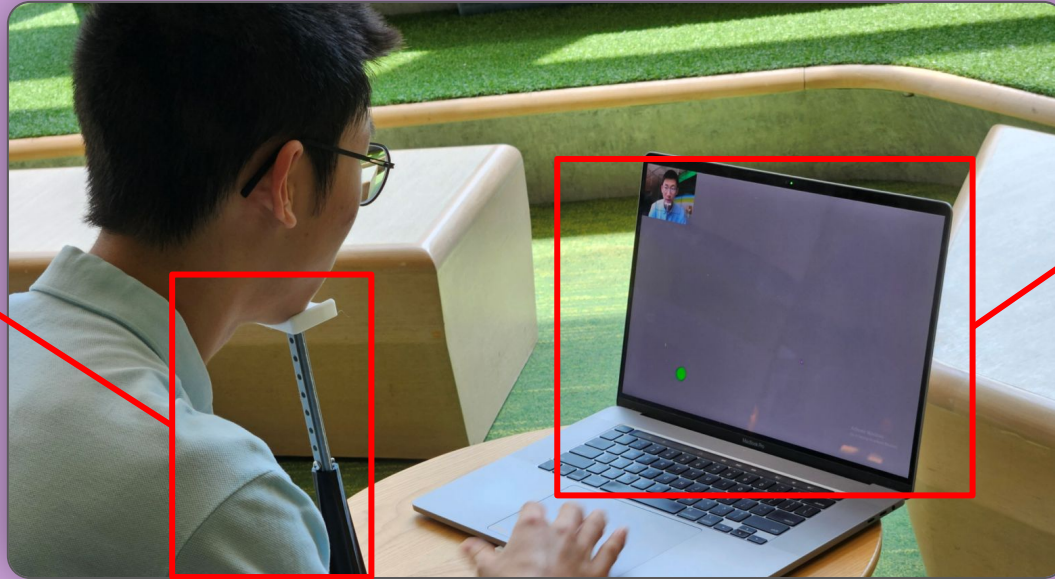


ALICE

Alzheimer Diagnosis and Severity Indicator based on Eye Behaviour Neural Network Classifier

ALEX MK1

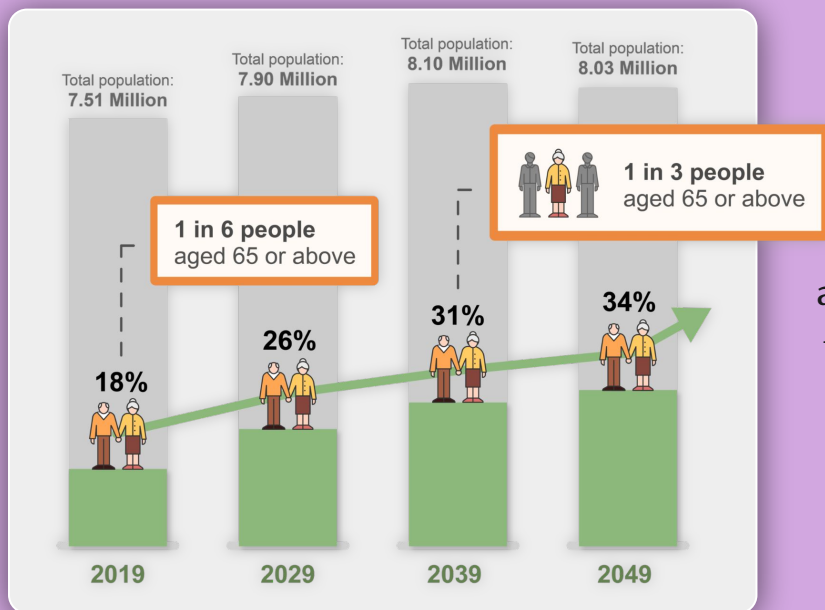


ALICE
(Data collection
utility shown)

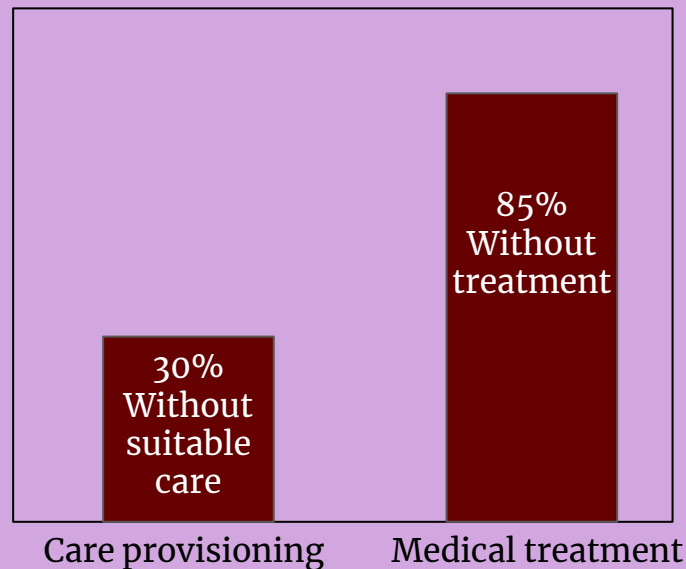
Designed & Created by:
YUQI SUN, KENNETH LAU, DAYU XU

Problem

Aging population in HK with Alzheimer's
Highly inefficient and wasteful care provision



Elderly
affected with
Alzheimer's



*Data from HKSAR
Government

Our solution: ALICE

Alzheimer Diagnosis and Severity Indicator based on
Eye Behaviour Neural Network Classifier



These are photos of the actual
product, not a 3D render

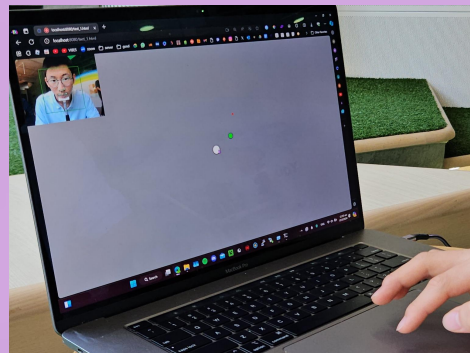
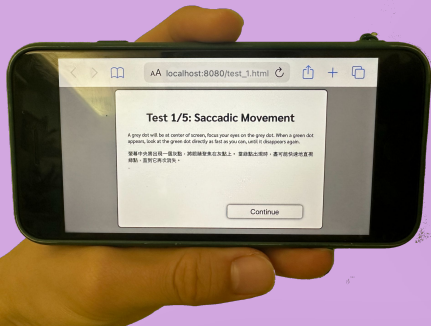
ALICE - Benefits



Accurate

Organizations can base treatments on the data to give precise aid quickly

1. Doctors see the effectiveness of treatment/prevention
2. Caregivers see the right amount of care needed
3. No over/under allocation of care, reduces inefficiency



Accessible

Most accessible service aimed to help everyone in need

1. Publicly available through internet to reach a wider audience
2. Accessible with minimal equipment, a webcam and internet
3. Remote access to conduct diagnosis anywhere to provide maximum coverage



Accelerated

Elderly can frequently take the test for most timely data

1. Tracks the progression of their conditions
2. Provide doctors and caregivers most timely data
3. Understands their current situation to give effective care

ALICE - Scientific Principles

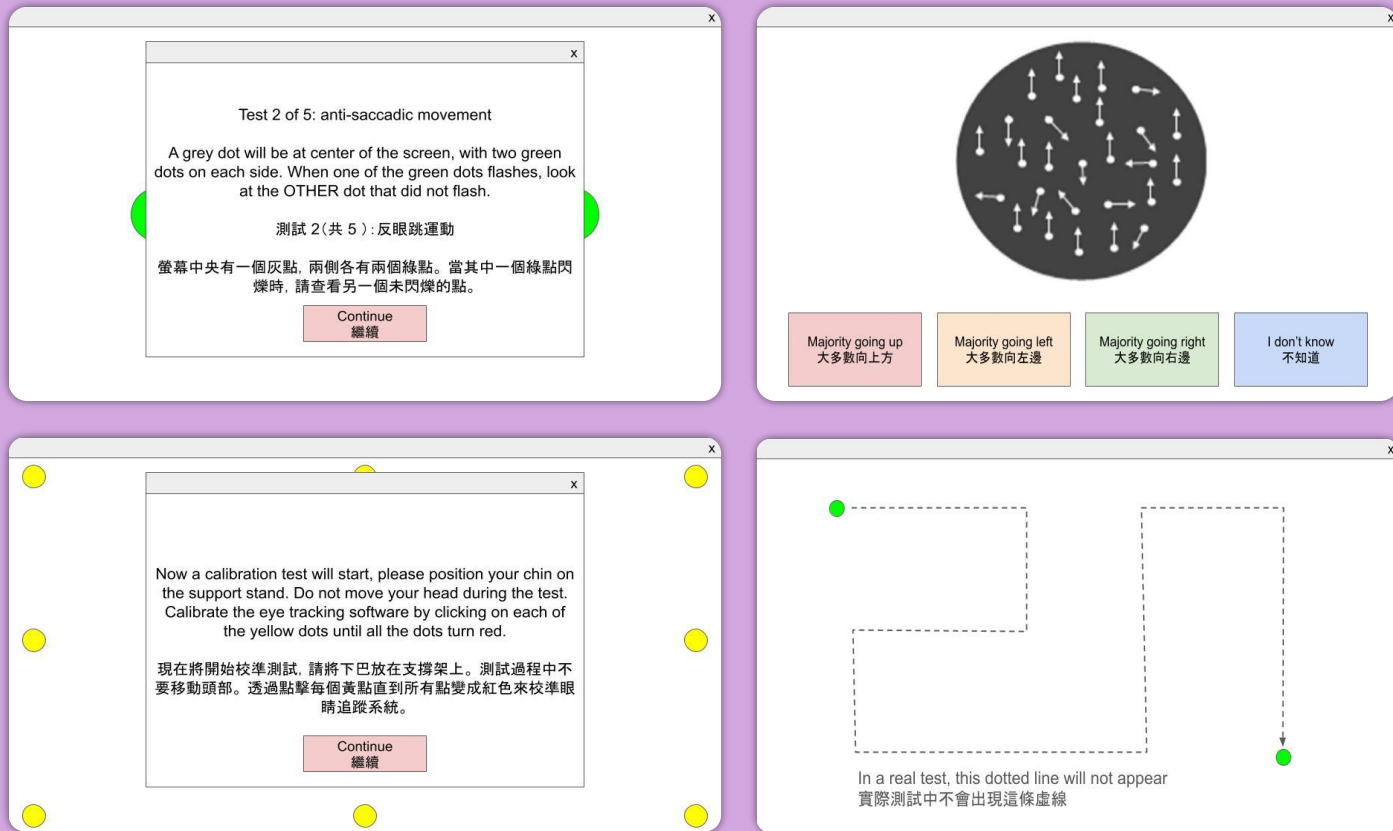


Fig 1-4 . ALICE Diagnostic website interface

Development Cycle

Collect 400 patients using ALICE:

- 150 control group
- 250 Alzheimer's patients of varying severity

Label the severity of patients by cross-checking other verified sources

Train the RNN to recognize patients' eye behavior with the labeled severity

The trained RNN model can classify severity of unseen data

Operational

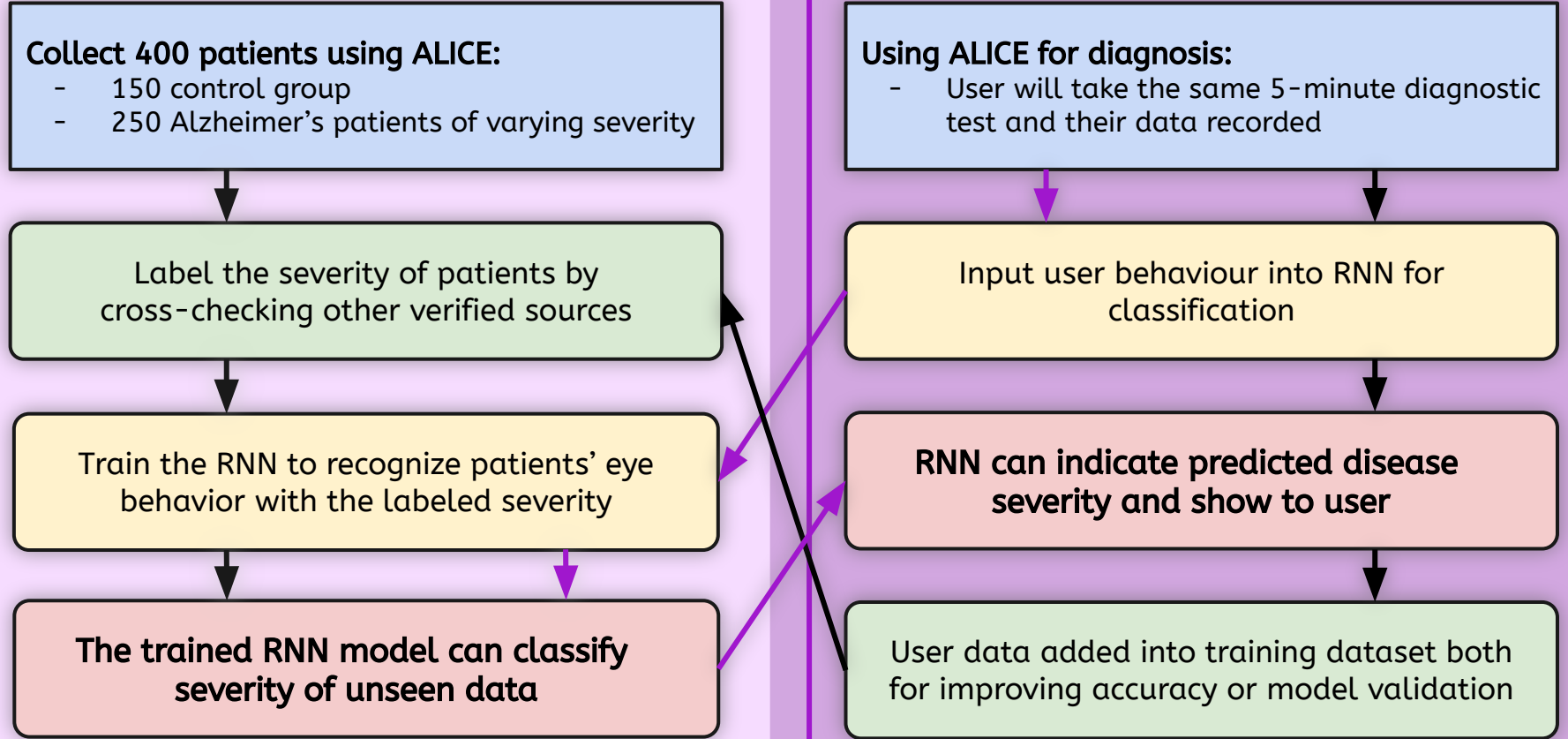
Using ALICE for diagnosis:

- User will take the same 5-minute diagnostic test and their data recorded

Input user behaviour into RNN for classification

RNN can indicate predicted disease severity and show to user

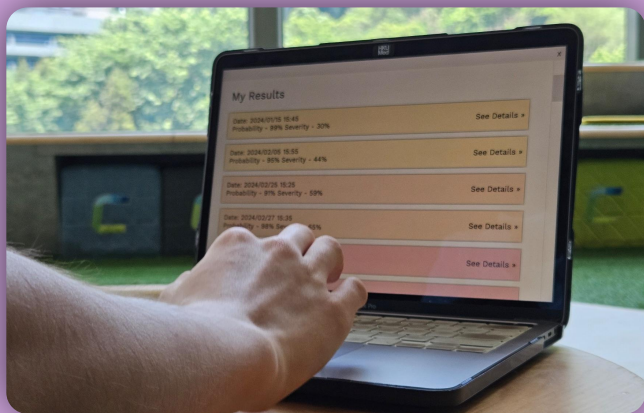
User data added into training dataset both for improving accuracy or model validation



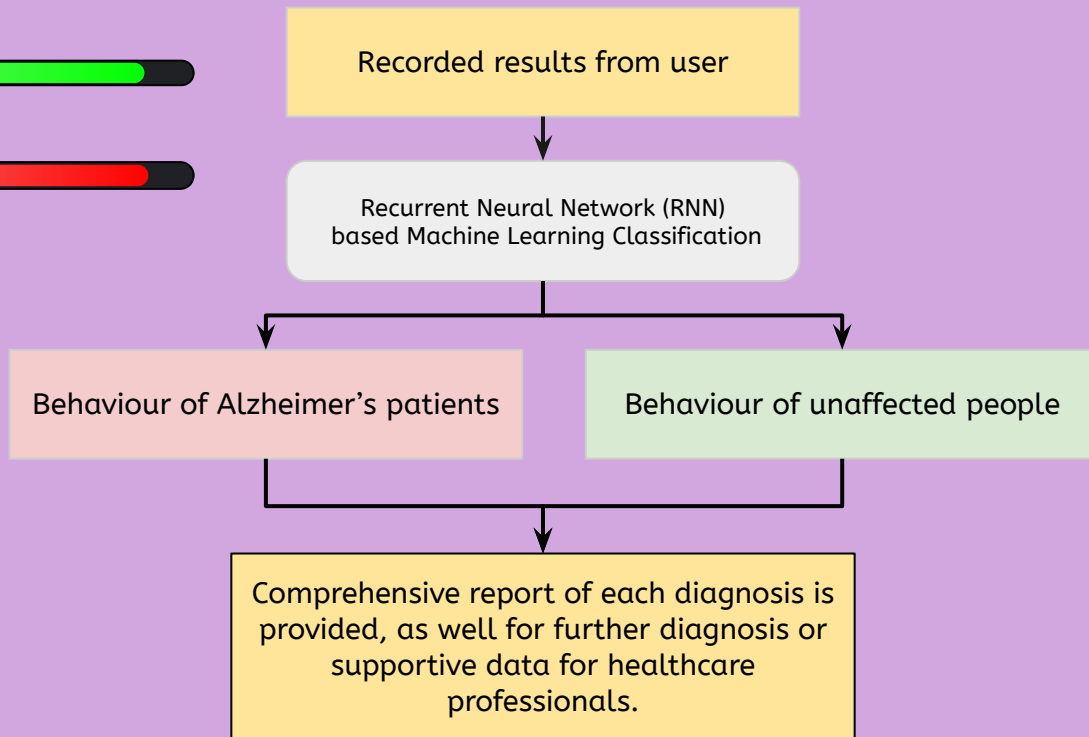
ALICE - Innovative & Patentable Technology

Probability: **90%** 

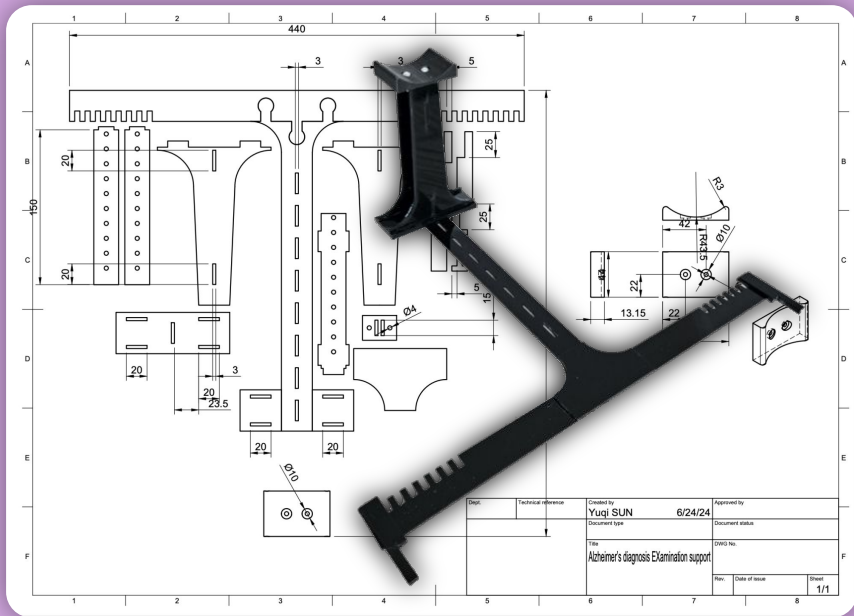
Severity: **90%** 



Example of results page after diagnosis



ALICE & ALEX - Plan for Production



ALEX Production Blueprints: [link](#)

v0.1.1

Latest

Combined all the testing pages into one file where web gazer can run continuously for data collection utility future improvements:

1. replace web gazer with recording instead
2. send recording to server for processing SAAS
3. use coordinate system of a virtual projected hypothetical plane instead of screen
4. data collection utility save results local or send to cloud

▼ Assets 2

Source code (zip)

2 week

Source code (tar.gz)

2 week

Software Distribution through GitHub

ALICE - Plan for User Trial



1. Readily large amount of users for testing
2. Alzheimer Patients available for testing
3. Unaffected individuals available for testing

2024 Jul

Completion of Initial Patient Data Collection

- Finish collecting $n=250$ Alzheimer's patient data
- Finish collecting $n=150$ Unaffected control group

2024 Oct

MVP Launch for Open Beta Testing

- ALICE is available for free personal usage
- Expand collaboration with more Universities & Hospitals (Currently 1 university and 1 hospital in HK)

2025 Jun

ALICE Full Release

- ALICE is open to the public with its full featureset
- ALICE free trial for commercial and enterprise usage to collect user data and feedback

ALICE - Long-Term Development

